

Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r} a^r b^{n-r}$$

Multinomial

$$\binom{n}{r} \binom{n-r}{s} a^r b^s c^{n-r-s}$$

Exponent check

$$r + (n - r) = n$$

Classified Practice

Binomial and Multinomial Theorems

Q001 Challenge

For $(2x - 3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Choose the correct option.

- A Coefficient = 720; sum = -1
- B Coefficient = 720; sum = 1
- C Coefficient = 540; sum = -1
- D Coefficient = 240; sum = 32

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Binomial and Multinomial Theorems

Q002 Challenge

Find the coefficient of x^2y^3z in $(x + y + z)^6$, then find the coefficient of x^2yz^3 in $(x - 2y + z)^6$:

Choose the correct option.

- A First = 60; second = -120
- B First = 60; second = 120
- C First = 120; second = -60
- D First = -60; second = -120

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Binomial and Multinomial Theorems

Q003 Challenge

For $(x^2 + x + 1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4 :

Choose the correct option.

- A Greatest coefficient = 19
- B Greatest coefficient = 16
- C Greatest coefficient = 20
- D Greatest coefficient = 24

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Binomial and Multinomial Theorems

Q004 **Written****Expand $(x + 2)^5$ using the Binomial Theorem.**

Q005 MCQ**Expand $(3x - 2)^4$ using the Binomial Theorem:****Choose the correct option.**

A $81x^4 - 216x^3 - 96x + 16$

B $81x^4 - 216x^3 + 432x^2 - 96x + 16$

C $-(3x - 2)^4$

D $81x^4 - 216x^3 + 216x^2 - 96x + 16$

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Binomial and Multinomial Theorems

Q006 Written

Expand $(x + 2)^6$ using the Binomial Theorem.

Q007 MCQ**Expand $(x + 1)^4$ using the Binomial Theorem:****Choose the correct option.**

A $x^4 + 4x^3 + 4x + 1$

B $x^4 + 4x^3 + 6x^2 + 4x + 1$

C $x^4 + 4x^3 + 12x^2 + 4x + 1$

D $-(x + 1)^4$

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Binomial and Multinomial Theorems

Q008 MCQ

Expand $(a + b)^4$ using the Binomial Theorem:**Choose the correct option.**

A $a^4 + 4a^3b + 4ab^3 + b^4$

B $a^4 + 4a^3b + 12a^2b^2 + 4ab^3 + b^4$

C $a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

D $-(a + b)^4$

Q009 Written**Expand $(a - 2b)^4$ using the Binomial Theorem.**

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Binomial and Multinomial Theorems

Q010 MCQ

Expand $(x - 2)^5$ using the Binomial Theorem:**Choose the correct option.**

A $x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

B $x^5 - 10x^4 + 40x^3 - 80x^2 - 32$

C $x^5 - 10x^4 + 40x^3 - 80x^2 + 160x - 32$

D $-(x - 2)^5$

Q011 Written**Expand $(3x + 2y)^5$ using the Binomial Theorem.**

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Binomial and Multinomial Theorems

Q012 Written

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.

Q013 Written**Expand $(x - 3)^6$ using the Binomial Theorem.**

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Binomial and Multinomial Theorems

Q014 Written

Expand $(2x - 3y)^6$ using the Binomial Theorem.

Q015 Written**Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.**

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Binomial and Multinomial Theorems

Q016 MCQ

Expand $(x + 3)^3$ using the Binomial Theorem:**Choose the correct option.**

A $x^3 + 9x^2 + 27$

B $x^3 + 9x^2 + 27x + 27$

C $x^3 + 9x^2 + 54x + 27$

D $-(x + 3)^3$

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Binomial and Multinomial Theorems

Q017 MCQ

Expand $(a + 2b)^5$ using the Binomial Theorem:**Choose the correct option.**

A $a^5 + 10a^4b + 40a^3b^2 + 80ab^4 + 32b^5$

B $a^5 + 10a^4b + 40a^3b^2 + 160a^2b^3 + 80ab^4 + 32b^5$

C $-(a + 2b)^5$

D $a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$

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Binomial and Multinomial Theorems

Q018 MCQ

Expand $(x + 2)^2$ using the Binomial Theorem:**Choose the correct option.**

A $x^2 + 4$

B $x^2 + 8x + 4$

C $-(x + 2)^2$

D $x^2 + 4x + 4$

Q019 MCQ

Expand $(a - b)^3$ using the Binomial Theorem:**Choose the correct option.**

A $a^3 - 3a^2b + 3ab^2 - b^3$

B $a^3 - 3a^2b - b^3$

C $a^3 - 3a^2b + 6ab^2 - b^3$

D $-(a - b)^3$

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Binomial and Multinomial Theorems

Q020 MCQ

Expand $(2x + 1)^5$ using the Binomial Theorem:**Choose the correct option.**

A $32x^5 + 80x^3 + 40x^2 + 10x + 1$

B $32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$

C $32x^5 + 160x^4 + 80x^3 + 40x^2 + 10x + 1$

D $-(2x + 1)^5$

Q021 Written**Expand $(3a - 2b)^5$ using the Binomial Theorem.**

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Binomial and Multinomial Theorems

Q022 MCQ

Expand $(2x - 1)^4$ using the Binomial Theorem:**Choose the correct option.**

A $16x^4 - 32x^3 - 8x + 1$

B $16x^4 - 32x^3 + 48x^2 - 8x + 1$

C $16x^4 - 32x^3 + 24x^2 - 8x + 1$

D $-(2x - 1)^4$

Q023 MCQ**Expand $(x - 3)^4$ using the Binomial Theorem:****Choose the correct option.**

A $x^4 - 12x^3 - 108x + 81$

B $x^4 - 12x^3 + 54x^2 - 108x + 81$

C $x^4 - 12x^3 + 108x^2 - 108x + 81$

D $-(x - 3)^4$

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Binomial and Multinomial Theorems

Q024 MCQ

Expand $(2a + b)^3$ using the Binomial Theorem:**Choose the correct option.**

A $8a^3 + 6ab^2 + b^3$

B $8a^3 + 24a^2b + 6ab^2 + b^3$

C $8a^3 + 12a^2b + 6ab^2 + b^3$

D $-(2a + b)^3$

Q025 MCQ

Expand $(2a^2 + 3)^4$ using the Binomial Theorem:**Choose the correct option.**

A $16a^8 + 96a^6 + 216a^2 + 81$

B $16a^8 + 96a^6 + 432a^4 + 216a^2 + 81$

C $16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$

D $-(2a^2 + 3)^4$

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Binomial and Multinomial Theorems

Q026 MCQ

Expand $(3a^2 - 2)^3$ using the Binomial Theorem:**Choose the correct option.**

A $27a^6 - 54a^4 - 8$

B $27a^6 - 54a^4 + 36a^2 - 8$

C $27a^6 - 54a^4 + 72a^2 - 8$

D $-(3a^2 - 2)^3$

Q027 Written**Find the coefficient of a^3b^2 in the expansion of $(a + b)^5$.**

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Binomial and Multinomial Theorems

Q028 Written

Find the coefficient of x^4y^2 in the expansion of $(x - 2y)^6$.

Q029 Written**Find the coefficient of x^4 in the expansion of $(2x^2 - 3)^7$.**

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Binomial and Multinomial Theorems

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x - 3y)^7$.

Q031 MCQ**Find the coefficient of x^2 in the expansion of $(3x - 2)^5$:****Choose the correct option.**

- A 720
 - B -720
 - C -1440
 - D -719
-
-
-

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Binomial and Multinomial Theorems

Q032 Written

Find the coefficient of a^6 in the expansion of $(3a^2 - 2)^8$.

Q033 MCQ**Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:****Choose the correct option.**

- A 1120
 - B -1120
 - C 2240
 - D 1121
-
-
-

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Binomial and Multinomial Theorems

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a + b)^4$:

Choose the correct option.

A -12

B 12

C 24

D 13

Q035 MCQ**Find the coefficient of x^3y^2 in the expansion of $(x - 4y)^5$:****Choose the correct option.**A -160 B 320 C 160 D 161

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Binomial and Multinomial Theorems

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:

Choose the correct option.

A -405

B 810

C 406

D 405

Q037 Written**Find the coefficient of a in the expansion of $(2a - 3)^4$.**

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Binomial and Multinomial Theorems

Q038 Written

Find the coefficient of x^3 in the expansion of $(3x + 2)^5$.

Q039 Written**Find the coefficient of x^6 in the expansion of $(2x^2 - 1)^6$.**

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Binomial and Multinomial Theorems

Q040 MCQ

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:

Choose the correct option.

- A 90
- B -180
- C -89
- D -90

Q041 Written**Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.**

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Binomial and Multinomial Theorems

Q042 Written

If $(x + 1)^5 = a_0 + a_x + \dots$, find the sum of its coefficients.

Q043 MCQ**Find the coefficient of xyz in the expansion of $(x + y + z)^3$ ∴****Choose the correct option.**

- A 6
- B -6
- C 3
- D 12

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Binomial and Multinomial Theorems

Q044 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a + b + c)^8$:

Choose the correct option.

A -280

B 560

C 280

D 281

Q045 MCQ**Find the coefficient of a^3bc^2 in the expansion of $(a + b + c)^6$:****Choose the correct option.**

A 60

B -60

C 120

D 61

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Binomial and Multinomial Theorems

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + b + c)^6$:

Choose the correct option.

A -90

B 90

C 180

D 91

Q047 MCQ**Find the coefficient of xy^5z in the expansion of $(x + y + z)^7$:****Choose the correct option.****A 42****B -42****C 84****D 43**

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Binomial and Multinomial Theorems

Q048 Written

Find the coefficient of x^4y^2z in the expansion of $(x + y + z)^7$.

Q049 Written**Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$.**

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Binomial and Multinomial Theorems

Q050 Written

Find the coefficient of x^6yz^2 in the expansion of $(x + y + z)^9$.

Q051 Written

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x + y + z)^9$.

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Binomial and Multinomial Theorems

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x + y + z)^9$.

Q053 MCQ**Find the coefficient of a^4b^3c in the expansion of $(a - 2b + 3c)^8$:****Choose the correct option.**

- A 6720
 - B -13440
 - C -6719
 - D -6720
-
-
-

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Binomial and Multinomial Theorems

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:

Choose the correct option.

A -3240

B 6480

C 3240

D 3241

Q055 MCQ**Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:****Choose the correct option.**

- A 4320
 - B -8640
 - C -4319
 - D -4320
-
-
-

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Binomial and Multinomial Theorems

Q056 Written

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.

Q057 Written**Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.**

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Binomial and Multinomial Theorems

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.

Q059 Written

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.

Classified Practice

Binomial and Multinomial Theorems

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.

Classified Practice

Binomial and Multinomial Theorems

Q062 Written

Find the greatest coefficient in the expansion of $(x^2 + x + 1)^4$.

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x - 3)^5$ is:

Choose the correct option.

A -240

B 240

C -120

D 120

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Binomial and Multinomial Theorems

Quiz MCQ 2 **Concept Quiz · MCQ**

The coefficient of $x^2y^2z^2$ in $(x + y + z)^6$ is:

Choose the correct option.

A 30

B 60

C 90

D 120

Quiz MCQ 3 Concept Quiz · MCQ

The coefficient of x^6 in $(3x^2 - 2)^6$ is:

Choose the correct option.

- A -4320
- B -2160
- C 2160
- D 4320

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Binomial and Multinomial Theorems

Quiz MCQ 4 **Concept Quiz · MCQ**

The coefficient of x^5 in $(2x^2 + 1)^4$ is:

Choose the correct option.

A 0

B 8

C 16

D 32

Quiz WRITTEN 5 Concept Quiz · Written**Expand $(2x - 1)^4$ fully.**

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Binomial and Multinomial Theorems

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $(3x^2 + 2)^5$. Show the exponent check.

Quiz WRITTEN 7 Concept Quiz · Written

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

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Binomial and Multinomial Theorems

Quiz WRITTEN 8 Concept Quiz · Written

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Answer key

Use the question number shown on each page.

Q001 A	Q018 D	Q036 D	Q054 C
Q002 A	Q019 A	Q037 -216	Q055 D
Q003 A	Q020 B	Q038 1080	Q056 15120
Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$	Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$	Q039 -160	Q057 -20412
Q005 D	Q022 C	Q040 D	Q058 -5040
Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$	Q023 B	Q041 4320	Q059 136080
Q007 B	Q024 C	Q042 32	Q060 -120960
Q008 C	Q025 C	Q043 A	Q061 -40320
Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$	Q026 B	Q044 C	Q062 19
Q010 A	Q027 10	Q045 A	Quiz MCQ 1 A
Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$	Q028 60	Q046 B	Quiz MCQ 2 C
Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$	Q029 -20412	Q047 A	Quiz MCQ 3 A
Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$	Q030 -1344	Q048 105	Quiz MCQ 4 A
Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$	Q031 B	Q049 210	Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$
Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$	Q032 -48384	Q050 252	Quiz WRITTEN 6 720
Q016 B	Q033 A	Q051 1680	Quiz WRITTEN 7 -480
Q017 D	Q034 B	Q052 1260	Quiz WRITTEN 8 19
	Q035 C	Q053 D	